

EquiCare — Sensor requirements for steps, lameness, watering and feeding

To: Sparsh CCTV

From: Bharat Sports Venture — EquiCare engineering

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Covers the remaining client monitoring points: Steps/locomotion · Lameness/limb-favoring · Watering (quantity/pattern/time) · Feeding (quantity/pattern/time)

1. Background

The thermal + colour camera you supplied covers **8 of the client's 12 monitoring points**: body temperature, respiration pattern, respiratory rate, general activity/abnormalities, resting pattern and duration, stable vices (weaving), urination pattern and excretion pattern.

This document covers the **remaining four points**. For each point it states what we need to measure, **how many sensors**, and **exactly where each sensor is placed**. We would like Sparsh to supply these sensors, or to source them for us.

All analytics are done by our software — step counting, gait and lameness analysis, intake patterns and alerts. From the sensors we need **raw, timestamped data** and a **documented way to receive it**. We are not buying analytics, apps or a dashboard.

The sensors worn by the horse must send data **live, including when the horse is outside the barn** for an hour or more of exercise. For that, one of the horse's devices carries a **mobile SIM** (section 5).

2. Summary — what goes on each horse and in each stall

On each horse (3 devices):

Device	Qty	Exact position	Worn	Points served
A. Leg tag (motion sensor, IMU)	1	Outside (lateral) face of the left front cannon bone , midway between knee and fetlock — in a soft padded sensor strap	24/7	Steps; lameness (stride timing); limb-favoring of that leg
B. Halter hub with SIM (motion sensor + LTE-M / NB-IoT modem)	1	Centred on the crown piece of the halter , just behind the ears (the poll) — on a clip-on bracket that moves to the bridle's crown piece for ridden work	24/7	Live data for all three devices ; lameness (head movement)
C. Pelvis sensor (motion sensor, IMU)	1	On the midline of the croup , over the sacrum, between the two points of the croup — snapped onto a base pad taped to the coat	Exercise and trot-ups	Lameness (hind legs)

In each stall (3 fixed sensors):

Device	Qty	Exact position	Points served
D. Water meter — flow meter on the automatic drinker's supply pipe (or a weighed bucket where buckets are used)	1	On the supply pipe just before the stall's automatic drinker , outside the horse's reach	Watering
E. Weigh-back feeder — load cell under the feed bowl	1	Feed bowl at the stall front , rim at chest height (0.9–1.1 m); hopper and electronics on the aisle side of the wall	Feeding (concentrates)
F. Hay load cell	1	In the hanging point of the hay net or hay rack (ceiling beam or wall hook)	Feeding (hay / forage)

All three horse devices must be supplied already fitted in their mounts (section 9): nothing is invasive — no clipping, no needles, nothing implanted.

Shared per site: our on-site computer (edge box, NVIDIA Jetson) — supplied by us. The stall sensors (D, E, F) connect to it by cable; the horse's sensors (A, B, C) reach our server over the mobile network through the hub.

3. Requirements common to all sensors

#	Requirement	Target
C1	Documented, open protocol , with the protocol document / register map sent with your reply	Modbus RTU/TCP (RS-485), pulse, MQTT or HTTPS; BLE GATT between horse devices
C2	No mandatory vendor cloud in the data path	Data reaches our server / edge box only; data stays in India (DPDP-2023)
C3	Raw data available, not only summaries	Per-axis motion waveforms; per-event water and feed amounts
C4	Timestamps on the device	Clock synchronised by NTP / network time; state drift over 24 h
C5	Buffering when the link is lost	≥24 h , replayed with the original timestamps
C6	Per-reading status / quality flag	e.g. valid, saturated, in motion, sensor fault
C7	Unique, fixed hardware ID in every record	Bindable to one horse / one stall through the API
C8	India approvals	WPC/ETA for every radio, TEC (MTCTE) for the SIM device, BIS

#	Requirement	Target
C9	Environment	IP67 for horse-worn devices, IP66/67 for stall devices; 0–50 °C; monsoon humidity; daily wash-down; ammonia
C10	Any software for our on-site computer	Built and tested for Linux ARM64 (NVIDIA Jetson) — or an open protocol that needs none

4. Point 1 — Steps / locomotion

4.1 What we will measure

Steps per minute, hour and day; walking, trotting and cantering time; movement in the stall at night — **24/7**, in the stall **and** outside it during exercise.

4.2 Sensors needed

1 leg tag (device A) per horse. The same tag also serves lameness (section 6).

4.3 Exact placement

- **Left front leg**, on the **outside (lateral) face of the cannon bone**, midway between the knee and the fetlock joint.
- Supplied **fitted in a soft padded sensor strap** (section 9.1): the sensor sits in a closed pocket on the outside of the leg, and the strap closes with hook-and-loop. It must sit flat and not rotate around the leg; strap and case show which way is **up** and which way is **forward**.
- **The same leg on the same horse every day**, so that the data stays comparable.
- Removable in seconds for grooming, bandaging and charging.

4.4 Live data

Leg tag → (Bluetooth, on the horse) → **halter hub with SIM** → (mobile network) → our server. See section 5.

4.5 Requirements

#	Requirement	Target
S1	Motion sensor	6-axis minimum (3-axis accelerometer + 3-axis gyroscope); 9-axis preferred. Part number + datasheet
S2	Accelerometer range	$\geq \pm 16 \text{ g}$ (kicks against the stall wall must not clip)
S3	Gyroscope range	$\geq \pm 2000 \text{ }^\circ/\text{s}$
S4	Sample rate	$\geq 100 \text{ Hz}$, configurable remotely

#	Requirement	Target
S5	Raw per-axis data	Available to us; any on-tag step counting or activity classification can be switched off
S6	Step counting (if you provide it)	State whether it is validated on horses , and its error at walk and trot
S7	Weight and size	≤50 g ; state dimensions
S8	Battery	≥7 days at 100 Hz, or rechargeable with a hot-swap spare so no data is lost
S9	Mount	Supplied in the padded sensor strap of section 9.1, sizes pony to draft horse
S10	Detach alert	Hub reports immediately when the tag goes silent or comes off
S11	Protection	IP67 or better; survives mud, urine, manure, sweat and hosing

5. Live data — the halter hub with SIM

5.1 Why

When the horse leaves the barn for exercise (arena, lunging, rides), Bluetooth loses the barn. We have decided on a **device with a mobile SIM** so that the horse stays monitored **live** wherever there is mobile coverage.

5.2 Sensors needed

1 halter hub (device B) per horse. It contains its **own motion sensor** (used for lameness, section 6), a **mobile modem with SIM**, and a larger battery. The leg tag and pelvis sensor stay small and light because the hub does the long-range sending for them.

5.3 Exact placement

- Centred on the **crown piece of the halter**, just behind the ears (the poll), on a **low-profile clip-on bracket** (section 9.2) — rigid, so the hub cannot turn or swing (its head measurements depend on that).
- For ridden work the horse wears a bridle: the hub must **move to the bridle's crown piece in seconds** on the same bracket, and sit in the same position.
- For 24/7 wear the halter must be a **breakaway halter** (it releases if the horse catches it).
- It must not press on the poll or rub, and must not catch on stall fittings.

5.4 What it does

- Collects the leg tag's and pelvis sensor's data over Bluetooth, and adds its own.
- Sends to **our server** over **LTE-M and/or NB-IoT**, with a fallback (e.g. LTE Cat-1) where those are not available.

- Sends **summaries at least once a minute** while the horse is active (steps, activity, lying/standing, and per-stride measures during trot), and **alerts immediately** (device removed, a sensor silent).
- Delivers the **raw waveforms** afterwards — over the mobile link, or offloaded over Bluetooth / Wi-Fi when the horse is back in the barn.
- Keeps everything it cannot send (**≥24 h**) and fills gaps later with the original timestamps.

5.5 Requirements

#	Requirement	Target
H1	Mobile modem	LTE-M and/or NB-IoT , with a fallback; module part number; networks and bands tested in India
H2	Sending to our server	MQTT or HTTPS over TLS to an address we configure ; documented payload with field names and units (sample attached)
H3	SIM	Physical SIM or eSIM; we prefer to choose the operator / use our own SIM
H4	Mobile data	State MB per horse per day — in the stall, and during 1 h of exercise
H5	Reporting	Summaries ≤1 min apart; alerts immediately; state end-to-end delay
H6	Its own motion sensor	Same as the leg tag (S1–S3), sample rate 200 Hz preferred (head movement at trot)
H7	Weight	≤150 g (our target — state yours)
H8	Battery	≥7 days with summaries every minute (our target — state actual at 1-min and 5-min reporting); power-saving (PSM / eDRX)
H9	Location (optional)	GPS/GNSS for position, speed and distance on rides — can be switched off
H10	Updates	Firmware and settings updated remotely over the mobile link
H11	Approvals	TEC (MTCTE), WPC/ETA, BIS
H12	Exercise scenario	The horse is out 1–2 hours with patchy coverage: state exactly what is sent live, what is stored, how gaps are filled, and whether anything can be lost

6. Point 2 — Lameness / limb-favoring

6.1 How we will measure it

Our software measures lameness while the horse **trots in a straight line on firm ground** (a vet's trot-up, or straight stretches during exercise):

- A horse lame in a **front leg nods its head** — the head rises as the sore leg lands. The **head sensor** (hub) measures this.
- A horse lame in a **hind leg hikes its hip** on the sore side. The **pelvis sensor** measures this.
- The **leg tag** gives the exact moment each stride lands, which tells us **left from right**.
- At rest in the stall, the leg tag also shows **limb-favoring** of its leg (pointing, resting, shifting weight off it).

Each horse is compared with its own normal; a change is flagged for a vet to examine. The sensors must therefore provide **clean, well-synchronised raw motion data** from all three positions.

6.2 Sensors needed

3 per horse, two of which are already listed:

- **Head:** the **halter hub** (device B) — no extra device.
- **Front leg:** the **leg tag** (device A) — no extra device.
- **Pelvis:** **1 pelvis sensor (device C)** — the only additional device. It is worn **during exercise and trot-ups**, not 24/7.

*The leg tag and the pelvis sensor can be the **same product** with a different mount.*

6.3 Exact placement

- **Head:** the hub, centred on the halter or bridle crown piece at the poll (section 5.3).
- **Pelvis:** on the **midline of the croup, over the sacrum, between the two points of the croup**. The sensor **snaps onto a thin base pad** that is stuck to the coat with **horse-safe double-sided tape** — no clipping — and peeled off after the session (section 9.3). It must stay in place at **canter** and not move on the skin.
- **Front leg:** the leg tag, as in section 4.3.

6.4 Requirements

#	Requirement	Target
L1	Motion sensor (pelvis)	Same as the leg tag (S1–S3)
L2	Sample rate (head and pelvis)	200 Hz preferred, 100 Hz minimum
L3	Timing between the three devices	Aligned within ≤10 ms of each other; state how
L4	Live summaries	The hub can send per-stride measures during trot, or forwards data for our server to compute
L5	Raw data	Full raw waveforms of each exercise session reach us, within hours at most
L6	Pelvis sensor battery	≥4 h of continuous recording per charge

#	Requirement	Target
L7	Pelvis mount	Supplied with the snap-on base pads and tape of section 9.3; stays in place at canter; no skin damage
L8	Quick pairing	The pelvis sensor can be paired to a different horse in under a minute (one sensor may be shared between horses exercised at different times)
L9	Validation data	Any equine gait or lameness data against a vet's lameness grading , if available

Out of scope for now: measuring the actual load on each leg (force plate).

7. Point 3 — Watering (quantity / pattern / time)

7.1 What we will measure

Every **drinking bout** — start time, end time and **millilitres** — the daily total, and the pattern across the day, **per horse** (one stall = one horse). A **refill** must be recorded separately from **drinking**.

7.2 Sensors needed

1 per stall, one of:

- **Option A (preferred): an inline flow meter** — electromagnetic or hall-effect — on the supply pipe of the stall's **automatic drinker**. The drinker refills as the horse drinks, so the water flowing in equals the water drunk (less small spills).
- **Option B: a weighed bucket** — a load cell in the bucket's wall bracket — for stalls that use buckets instead of an automatic drinker. Drinking shows as weight falling; refills as a jump upwards.

7.3 Exact placement

- **Option A:** on the incoming pipe **immediately before the drinker's valve**, on the **aisle side of the stall wall**, or inside a steel enclosure **above 1.8 m**. No part of the meter or its cable within the horse's reach.
- **Option B:** load cell in a steel wall bracket with the bucket rim at **0.9–1.1 m** (chest height); load cell and cable behind a steel guard.

7.4 Connection

Fixed in the stall, so **by cable** — RS-485 / Modbus (or pulse output) — to our on-site computer.

No SIM needed.

7.5 Requirements

#	Requirement	Target
W1	Measuring principle	Electromagnetic or hall-effect, no or few moving parts ; or load cell (option B)
W2	Resolution	≤10 ml per pulse / step (our target — state yours)

#	Requirement	Target
W3	Accuracy	$\pm 2-3$ % across 0.1–8 L/min (our target — state yours)
W4	Smallest flow measured	≤ 0.1 L/min — slow sips must not read as zero
W5	Bouts and refills	Each drinking bout timestamped with its volume; refills separate (option B)
W6	Water quality	Food-grade wetted parts; tolerant of hard Indian groundwater , scale and hay debris; state cleaning interval
W7	Output	Modbus RTU/TCP (RS-485) or pulse; register map attached
W8	Power	12–24 V DC or PoE; state watts
W9	Protection	IP66/67 ; corrosion-resistant; survives daily wash-down
W10	Load cell (option B)	Resolution ≤ 10 g, capacity ≥ 25 kg, rejects nudging and leaning; state drift over 24 h

8. Point 4 — Feeding (quantity / pattern / time)

8.1 What we will measure

For each meal: **grams offered**, **grams eaten** and **grams refused** (left over), with times, plus **hay eaten** across the day — the pattern and timing per horse. A horse leaving its feed is one of the earliest signs of illness, so **what was refused matters as much as what was given**.

8.2 Sensors needed

2 per stall:

- **1 weigh-back feeder (device E)** for concentrates (pellets, grain, mixes): it weighs what is in the bowl before and after each meal. An automatic dispenser with a per-horse schedule is preferred; a weighed bowl filled by hand is acceptable.
- **1 hay load cell (device F)** for forage, since horses eat mostly hay.

8.3 Exact placement

- **Feeder:** mounted at the **stall front or front corner**, bowl rim at **0.9–1.1 m** (chest height), with the **load cell under the bowl**. Hopper, motor and electronics on the **aisle side of the wall**, out of the horse's reach. The bowl must lift out for daily washing.
- **Hay:** load cell in the **hanging point of the hay net or hay rack** — the ceiling beam or wall hook where the net hangs today. Where hay is fed from the floor, a **weighed hay box** instead.

8.4 Connection

By cable — **RS-485 / Modbus** — to our on-site computer. **No SIM needed.**

8.5 Requirements

#	Requirement	Target
F1	Feed bowl weighing	Resolution ≤10 g ; accuracy ±20 g (our target — state yours)
F2	Weigh-back	Records offered and left over per meal, with times
F3	Dispensing (if automatic)	Per-horse schedule (morning / midday / evening) set through the local API ; each dispense recorded with scheduled vs actual time and grams
F4	Faults	Hopper empty, jam and motor stall reported as separate alarms
F5	Hay load cell	Capacity ≥20 kg , resolution ≤50 g ; separates eating from the net swinging
F6	Nudging and leaning	Short spikes (horse pushing the bowl or net) rejected without losing real eating
F7	Output	Modbus RTU/TCP (RS-485) ; register map attached
F8	Hygiene	Feed-contact parts food-grade, removable and disinfectant-safe without re-calibration
F9	Protection	IP66/67 ; kick, bite and lean resistant; electronics outside the stall
F10	Power	12–24 V DC or PoE; state watts

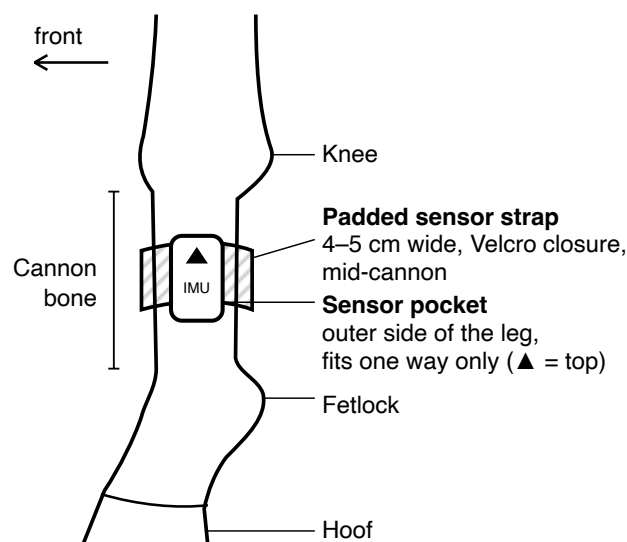
Until these sensors are installed, stable staff log feed and water in our software; the sensor data will use the same records.

9. Mounting — sensors supplied already fitted

We expect **each horse device delivered ready to wear, fitted in its mount**, with the mount designed together with the sensor. All mounts are **non-invasive**: no clipping, no needles, nothing implanted.

9.1 Leg tag — soft padded sensor strap

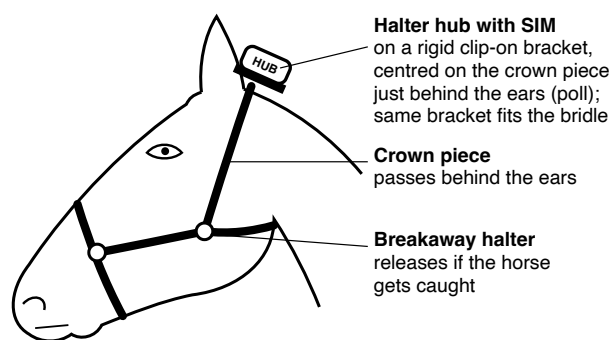
- A **padded strap about 4–5 cm wide** around the cannon bone, closed with **hook-and-loop (Velcro)**, with the sensor in a **closed pocket on the outside (lateral) face** of the leg.
- The pocket is **shaped so the sensor fits only one way**, and the strap is marked **top** and **front** — so the sensor sits at the same place and angle every time it is put on.
- Why a strap and not a bandage: bandages must be re-wrapped every 12–24 hours, uneven wrapping can injure the tendons, and each re-wrap moves the sensor. A shaped strap goes on the same way every time, in seconds.



Leg tag in a padded sensor strap — outer side of the left front cannon bone

9.2 Halter hub — clip-on bracket on the crown piece

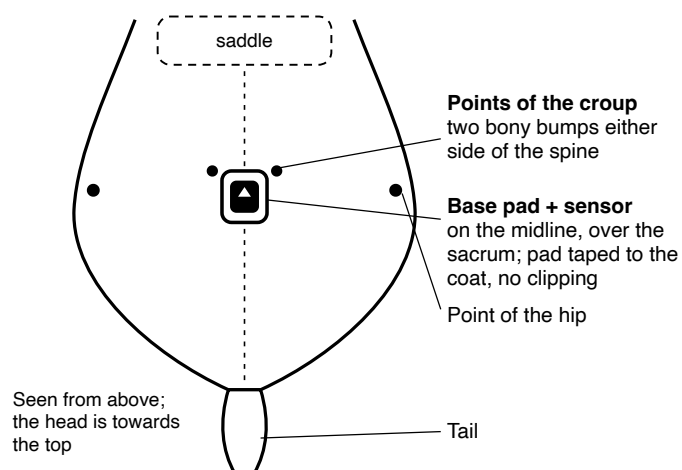
- A **rigid, low-profile bracket** that clips onto the crown piece of a standard halter **and** of a bridle, with a **quick-release** — so the same hub moves from halter to bridle in seconds.
- The hub sits **centred just behind the ears**, cannot turn or swing, and does not press on the poll.
- For 24/7 wear, a **breakaway halter** (supply one, or confirm the bracket fits common breakaway halters).



Halter hub with SIM on a clip-on bracket — crown piece, just behind the ears

9.3 Pelvis sensor — snap-on base pad

- A **thin, flexible base pad** stuck to the coat over the croup midline with **horse-safe double-sided tape**; the sensor **snaps onto the pad** and comes off in one movement.
- The pad is **single-use** or re-usable with fresh tape; it peels off after the session without pulling hair or marking the skin.



Pelvis sensor on a base pad — croup midline, over the sacrum

9.4 Requirements for all mounts

#	Requirement	Target
M1	Delivered fitted	Each sensor arrives in its mount , ready to put on the horse
M2	Sizes	Leg strap pony, horse, draft (or adjustable across that range); bracket fits standard halters and bridles
M3	Materials	Breathable, non-abrasive, hypoallergenic padding; no metal or hard edges against the skin
M4	Safety	Low profile, rounded edges , nothing that can catch on stall fittings; sensors on the outside of the leg only (the inside is hit by the other hoof)
M5	Orientation	Pocket / bracket / pad keyed so the sensor fits one way only; top and front marked
M6	Fitting time	Each device on or off in under 30 seconds by stable staff
M7	Hold	Stays in place through rolling, lying down, turnout and canter without slipping or rotating
M8	Cleaning	Straps machine-washable ; bracket and pad wipe-clean and disinfectant-safe
M9	Spares and wear	State strap service life; spare straps per horse so one can be washed; base pads and tape as consumables
M10	Fitting guide	Illustrated fitting instructions, including how tight (a finger's width under the strap)
M11	Welfare	Any evidence of 24/7 wear on horses without rubbing or sores (leg and halter)

10. What we need from Sparsh

1. **For each device (A–F):** datasheet, **the actual protocol document / register map**, and a **sample data file recorded from the exact model and firmware** you would supply.
2. **Your answer to every requirement row** above (C1–C10, S1–S11, H1–H12, L1–L9, W1–W10, F1–F10, M1–M11): the value you meet, or "**not supported**". A row left blank is recorded as not supported.
3. **India certificates** (WPC/ETA, TEC/MTCTE, BIS), or their status.
4. **Evaluation units for one horse and one stall**, horse devices **fitted in their mounts**: 3 leg tags in padded straps (with spare straps), 2 halter hubs with SIMs that work in India on clip-on brackets (plus a breakaway halter), 1–2 pelvis sensors with base pads and tape, 1 charging dock, 1 water meter (or weighed bucket), 1 weigh-back feeder, 1 hay load cell — with **availability and lead time**.
5. A **named technical contact** for clarifications.

Commercial terms are not part of this document — we will discuss them once the technical fit is confirmed.